

TECHNICAL REPORT ON PRODUCT DATASET

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HNG-14

INTRODUCTION

This report represents the data cleaning and preparation process carried out on a product dataset containing information such as product titles, descriptions, product id (unique identifier), product type id (category), and bullet points.

The dataset was reviewed to identify and address data quality issues, including missing values, duplicate records, and inconsistent formatting.

The objective of this task is to transform the raw dataset into clean and structured format suitable for analysis, while also creating a new feature, *short_title*, to improve product title readability and support search optimization.

All data cleaning and preparation processes were performed using Microsoft Excel.

DATA CLEANING PROCESS

During the exploration of the dataset, several data quality issues were identified. These issues included missing values in key columns such as *bullet_points*, *description*, *product_type_id*, and *product_length*. Additionally, duplicate records were observed, which could affect the accuracy of analysis. Inconsistent column naming formats were also identified, along with irregularities in the text data, such as overly long and unstructured product lines. To address these issues, appropriate data cleaning steps were carried out.

- Missing values in *bullet_points* and *description* columns were replaced with “Not Available” to maintain completeness.
- Missing entries in *product_type_id* columns were replaced with “Unknown”.
- Missing values in the *product_length* column were replaced using the average value of the column.
- Duplicate records (217 records) were removed using Excel’s built-in “Remove Duplicates” feature to ensure accuracy.
- Column names were standardized to a consistent lowercase format with underscores (e.g. *product_id*, *product_length*) to improve readability and consistency.
- Data types were also reviewed to ensure numerical fields were correctly formatted.

These steps resulted in a cleaner and more structured dataset suitable for further analysis.

SHORT TITLE CREATION

The creation of the *short_title* column was aimed at producing concise, readable and SEO-friendly product titles while retaining essential product information. However, several challenges were encountered due to inconsistent structure of the original product titles.

One major issue was the **variation in title length**, with many entries exceeding recommended 30–50-character range. Additionally, the titles contained **unnecessary words** such as “for”, “of”, “from”, which increased length without adding meaningful value. The presence of **special characters and delimiters** such as commas (,), pipes (|), hyphens (-), and symbols (&) also made it difficult to extract clean and uniform titles.

Another challenge was the **lack of consistent pattern** across the dataset. Some titles contained multiple descriptive segments separated by symbols, while others were continuous text. This inconsistency made it difficult to apply a single rigid extraction rule across all records.

To address these challenges, a combination of Excel text functions was applied in a structured manner.

- The **MID**, **LEFT** and **RIGHT** functions were used to extract relevant portions of the text based on character positions and delimiters.
- The **SUBSTITUTE** function was used to remove unwanted words (e.g., “for”, “of”, “from”) and eliminate unnecessary symbols.

- The **TRIM** function was applied to clean up extra spaces resulting from these transformations.
- **CONCATENATE** function was used to recombine meaningful parts of the text where needed.

The stepwise approach allowed for the systematic reduction of product titles while preserving key information such as product type, brand and important attributes. Although the process was largely automated using formulas, minor manual adjustments were made where necessary to ensure clarity and readability.

Overall, the resulting *short_title* column provides a more structured, concise and user-friendly representation of the original product titles, making the dataset more suitable for marketing and analytical purposes.

DATASET OVERVIEW

After completing the data cleaning and short title creation processes, the dataset was significantly improved in terms of structure, consistency and usability. Missing values across key columns such as *bullet_points*, *description*, *product_type_id*, and *product_length* were addressed, ensuring that the dataset no longer contained critical gaps that could affect analysis.

Duplicate records were identified and removed, resulting in a more accurate representation of data. In addition, column names were standardized to a consistent format, improving readability and making the dataset easier to work with. Data types were also reviewed and corrected where necessary, ensuring that numerical fields were properly formatted for analysis.

The introduction of the *short_title* column further enhanced the dataset by providing concise and structured product titles.

Compared to the original titles, the new short titles are more readable, consistent and suitable for marketing and analytical purposes. Unnecessary words and symbols were removed and title lengths reduced while retaining essential product information.

Overall, these improvements have made the dataset cleaner, more reliable and better prepared for further analysis and decision-making. However, some limitations remain, particularly due to the original inconsistency in product title structures, which required a combination of automated and minor manual adjustments.

A	B
1	PRODUCTID TITLE
2	1925202 ArtzFolio Tulip Flowers Blackout Curtain for Door, Window & Room Eyelets & Tie Back Canvas Fabric Width 4.5feet (54inch) Height 5 feet (60 inch); Set of 2 PCS
3	2673191 Marks & Spencer Girls' Pyjama Sets T86_2561C_Navy Mix_9-10Y
4	2765088 PRIKNIK Horn Red Electric Air Horn Compressor Interior Dual Tone Trumpet Loud Compatible with SX4
5	1594019 ALISHAH Women's Cotton Ankle Length Leggings Combo of 2, Plus 12 Colors_L
6	283658 The United Empire Loyalists: A Chronicle of the Great Migration
7	2152929 HINS Metal Bucket Shape Plant Pot for Indoor & Outdoor Gardening (Red, Medium) Plant Stands for Indoor Balcony Plant Bench Plant Stands Pot Stand Single Potted Plant Stand Big Pots Metal
8	413758 Ungifted: My Life and Journey
9	2026580 Delavala Self Adhesive Kitchen Backsplash Wallpaper, Oil Proof Aluminum Foil Kitchen Sticker (Sliver 5(Mtr))
10	2050239 PUMA Cali Sport Clean Women's Sneakers White Leather (37540701)
11	2998633 Hexwell Essential oil for Home Fragrance Oil Aroma Diffuser oil Set of 2 Rajnigandha Oil & TeaTree Oil -10ML Each
12	2857066 3NHÄ,Ä® Glasses Goggles Anti Fog Antis Windproof Anti Dust Resistant
13	833712 La Mure / Valbonnais gps
14	648364 J'ecris des lettres! premiers exercices d'ÄřÄ©criture 5-6 ans
15	1991694 Mediterranean diet for beginners: 7Benefits of mediterranean diet,7day plan and 70yummy easy recipes
16	2790448 SEGOVIA Single Walled Stainless Steel Sports fridge Water Bottle with SS Cap 1000 ml Bottle (straight+belly with sipper cap / cola+ smart with steel cap pack of 4 pcs)
17	1810976 Stone & Beam Fan Embossed Planter in Blue, Medium
18	1262926 Star Trek 50th Anniversary Cereamic Storage Jar
19	1491106 Steelbird Hi-Gn SBH-11 HUNK Glossy Black and Blue with Smoke visor,580 mm
20	1543564 Kenneth Cole REACTION Men's Crespo Loafer B Shoe, Cognac, 10 M US
21	793582 MASTER OF THE RINGS
22	1045826 Cvbtravd L049 No. 4 Lolly Chocolate Candv Mold. No. 4 with Exclusive Cvbtravd Coovrighted Chocolate Molding Instructions

Before data cleaning

A	B	
1	product_id short_title	bullet_points
2	1925202 Tulip Flowers Curtain_2 PCS	[LUXURIOUS & APPEALING: Beautiful custom-r
3	2673191 Girls' Pyjama T86_256_9_10Y	[Harry Potter Hedwig Pyjamas (6-16 Yrs),100% c
4	2765088 Electric Air Compress_h SX4	[Loud Dual Tone Trumpet Horn, Compatible Wi
5	1594019 Women's Cotton Length_ors L	[Made By 95%cotton and 5% Lycra which gives
6	283658 United Empire A Chron_ation	Not Available
7	2152929 Metal Bucket Plant Po_Metal	[Simple and elegant, great for displaying indoc
8	413758 My Life Journey_urney	Not Available
9	2026580 Kitchen Backsplash Oi_Mtr))	[HIGH QUALITY PVC MATERIAL: The kitchen alu
10	2050239 Women's Sneakers Leat_0701)	[Style Name:-Cali Sport Clean Women's Sneak
11	2998633 Home Fragrance Aroma_Each	[100% Pure And Natural Essential Oil Or Fragra
12	2857066 Glasses Goggles Fog A_stant	[Good quality and Suitable to use.,This Product
13	833712 Mure / gps_/gps	Not Available
14	648364 des lettres! exercice_6 ans	Not Available
15	1991694 diet beginners: medit_cipes	Not Available
16	2790448 Single Walled Steel S_pcs)	[Segovia bottle consists of stainless steel whic
17	1810976 Beam Fan Planter in B_edium	Not Available
18	1262926 Cereamic Storage Jar_e Jar	Not Available
19	1491106 Glossy Black Blue wit_80 mm	[High Impact ABS Material Shell,Replaceable ai
20	1543564 Men's Crespo B Shoe C_M US	[Burnished metal center bit,Slightly padded Fo
21	793582 MASTER OF RINGS_RINGS	Not Available
22	1045826 Chocolate Candv No 4_tions	Not Available

After data cleaning

E	F	G	H	I	J	K	L	E	F	G	H	I	J
1	PRODUCTTYPEID ProductLength							1	product_type_id product_length				
2	1650 2125.98							2	1650 2125.98				
3	2755 393.7							3	2755 393.70				
4	7537 748.0314953							4	7537 748.03				
5	2996 787.401574							5	2996 787.40				
6	6112 598.424							6	6112 598.42				
7	5725 950							7	5725 950.00				
8	23 598							8	23 598.00				
9	6030 984.2519675							9	6030 984.25				
10	3302 393.7							10	3302 393.70				
11	8201 393.700787							11	8201 393.70				
12	10359 590.5511805							12	10359 590.55				
13	1 433.07							13	1 433.07				
14	1 759.841							14	1 759.84				
15	32 600							15	32 600.00				
16	1273 314.9606296							16	1273 314.96				
17	6 670							17	6 670.00				
18	0 100							18	0 100.00				
19	8046 1240.155							19	8046 1240.16				
20	3247 500							20	3247 500.00				
21	716 750							21	716 750.00				
22	13101 1025							22	13101 1025.00				

Before data cleaning

After data cleaning

LIMITATIONS

Despite the improvements made, some limitations remain. The inconsistency in the original product titles made it difficult to fully automate the short title creation process. As a result, minor manual adjustments were required. Additionally, assumptions made during missing value handling may affect the interpretation of certain variables.